

Patient Multi-omics

- Gene Expression
- Mutated genes
- Mutation types
- Clinical Traits
- BeatAML trainc waves 1-2
- BeatAML test: waves 3-4



Gene expression
Mutated Genes
Mutation Types
Clinical Traits



Drugs

- Physicochemical features (PC)
- Embedding representation (Embed)



RDkit $\rightarrow$ PC

Transfer / $\rightarrow$ Embed

Zero-Shot
learning



Sources:

BeatAML drug, DrugBank, MOSES, ChEMBL, ChEMBLiRTs, MolFormer-XL

Existing Data

- Pathway Database
- StringPPI network
- BeatAML (Cell & Module states)



Pathway Genesets

PPI +drug targets

Cell +Module
Genesets



Pathway
enrichments

Cell +Module
Enrichment

Drug Pathway
AUC

Data

Feature engineering

Modeling

Analysis

Featured used

Models



SHAP Explanations

Patient Stratification



External Validation



SeeAML
FFNN-AML